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|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|           |                                                                                                                                               |           |
| 3(a)(i)   | $\sigma = F / xy$                                                                                                                             | <b>B1</b> |
| 3(a)(ii)  | $\varepsilon = (z - w) / w$                                                                                                                   | <b>B1</b> |
| 3(a)(iii) | $E = \sigma / \varepsilon$                                                                                                                    | <b>C1</b> |
|           | $= Fw / xy(z - w)$                                                                                                                            | <b>A1</b> |
| 3(b)(i)   | extension = 2.2 mm ( <i>allow</i> 2.0–2.4 mm)                                                                                                 | <b>A1</b> |
| 3(b)(ii)  | strain energy = area under graph/line <b>or</b> $\frac{1}{2}Fx$ <b>or</b> $\frac{1}{2}kx^2$                                                   | <b>C1</b> |
|           | $= \frac{1}{2} \times 120 \times 1.4 \times 10^{-3}$ <b>or</b> $\frac{1}{2} \times 8.6 \times 10^4 \times (1.4 \times 10^{-3})^2$             | <b>C1</b> |
|           | $= 0.084 \text{ J}$                                                                                                                           | <b>A1</b> |
| 3(b)(iii) | (some of the) deformation of the wire is plastic/permanent/not elastic<br><b>or</b><br>wire goes past the elastic limit/enters plastic region | <b>B1</b> |
|           | energy (that cannot be recovered) is dissipated as thermal energy/becomes internal energy                                                     | <b>B1</b> |